

Automated Detection of Brain Tumors in Magnetic Resonance Imaging (MRI) through Convolutional Neural Networks and Ensemble Learning

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Abstract—The early and accurate detection of brain tumors is fundamental for choosing the appropriate treatment and for clinical decision support. This study proposes an automated approach for the classification of brain tumors in magnetic resonance imaging (MRI), using pre-trained convolutional neural networks (VGG16, Xception, and InceptionResNetV2) and fine-tuning techniques, complemented by Ensemble learning and Grad-CAM for interpretation of predictions. The dataset used was the LGG MRI Segmentation, and the data were pre-processed with normalization, data augmentation, and class balancing. The individual models achieved good performances, with accuracy and AUC greater than 0.88 and 0.95, respectively, after fine-tuning. The Ensemble learning approach stood out with an AUC of 0.98, showing its generalization capacity. The application of Grad-CAM allowed visualizing the most relevant regions of the images for decision-making, increasing the reliability of the results. The results indicate that the proposed techniques are effective for tumor classification in MRI, with the potential to assist healthcare professionals in clinical decision-making, while also highlighting areas for future improvements, such as the reduction of false positives and adaptation to more complex cases.

Index Terms—Brain Tumor Detection, Medical Imaging, Magnetic Resonance Imaging (MRI) Classification, Convolutional Neural Networks (CNNs), Fine-Tuning, Ensemble learning, Explainable Artificial Intelligence (XAI), Clinical Decision Support

I. INTRODUCTION

The growing need for quick and accurate medical diagnoses for the effective treatment of cancer patients has driven the use of artificial intelligence techniques, particularly deep neural networks (Deep Learning), to automate some of the processes that were previously performed exclusively by healthcare professionals (medicine) [1].

The use of medical images, such as brain magnetic resonance images, has proven to be a powerful tool for detecting various neurological conditions, including brain tumors and lesions. Magnetic resonance imaging (MRI) is widely used to detect brain abnormalities, being a crucial tool for doctors and specialists in identifying tumors. However, the analysis of these images is often complex and time-consuming, as it requires a thorough visual assessment by qualified healthcare professionals, thus becoming a challenge for both doctors and

healthcare professionals, as well as for healthcare systems with a large volume of patients, which is what happens nowadays [2]. This challenge results in a high workload and a margin of human error, which can impact diagnostic effectiveness and response time in patient treatment.

The central problem addressed in this work is the need for automated and improved methods for analyzing brain magnetic resonance images for the classification/identification of brain tumors [2]. The accurate and efficient identification of these tumors is fundamental for choosing appropriate treatments and reducing the workload of healthcare professionals, who are completely overwhelmed with daily tasks, so the more automated these tasks are, the better for them and for all patients. Although there are traditional approaches based on conventional algorithms, they often lack the accuracy or certainty necessary to deal with the complexity of the problem, in this case, medical images, which can have various subtle variations and dispersed relevant information [3].

In this scenario, great opportunities arise for the application of Deep Learning models, such as convolutional neural networks (CNNs), which have shown impressive results in computer vision tasks, such as image classification in various areas, including medicine. The main advantage of these models lies in their ability to learn hierarchical data representations, allowing for a more detailed and accurate analysis of images, since they process large volumes of data and identify patterns that may go unnoticed by humans [1]. With the use of large volumes of data (Datasets) and the computational power available today, Deep Learning (DL) models can be trained to achieve higher performance levels than conventional methods, ultimately providing quick and accurate diagnoses, allowing for more agile decisions in patient treatment. Despite all this automated work, the healthcare professional will always have their role and will always make the final decision in any problem, because artificial intelligence will hardly replace that position.

The area of medical image classification, particularly those related to brain tumors or other types of serious diseases in magnetic resonance imaging (MRI) exams, has been widely investigated due to its importance in clinical diagnosis. Previ-

ous studies have addressed different approaches and methodologies to improve the detection and classification of diseases, ranging from methods based on manual feature extraction to more recent deep learning techniques.

For example, Noor et al. (2020) [4] analyzed 42 articles in their study, published between 2015 and 2019, focusing specifically on the application of deep learning methods to detect neurological diseases from magnetic resonance images. The main focus was on Alzheimer's, Parkinson's, and schizophrenia diseases. This work provides an overview of the use of DL for diagnosing these three specific neurological conditions.

Another relevant study was done by Litjens et al. (2017) [5], which analyzed a vast set of 300 articles published between 1995 and 2017, covering a wide range of applications of deep learning and machine learning in medical image analysis, including image/exam classification, object or lesion classification and detection, and lesion segmentation. This work offers an overview of the application of DL and ML for various image analysis tasks in general, including, but not limited to, the brain area.

In addition, Durstewitz et al. (2019) [6] conducted a review examining 65 articles published between 2008 and 2018, focusing on the application of deep neural networks in psychiatry, specifically addressing conditions such as schizophrenia, autism spectrum disorder, Parkinson's disease, depression, substance abuse disorder, and epilepsy. This study highlights the use of neural networks in the context of various neuropsychiatric diseases.

Finally, Erickson et al. (2017) [7] analyzed 85 articles, focusing on machine learning models and algorithms applied in medical imaging, with a special emphasis on Alzheimer's disease. This study focuses more on ML algorithms and models for medical applications.

TABLE I
SUMMARY OF LITERATURE REVIEW AND METHODOLOGIES

Ref.	Year	N° Art.	Period	Pathology / Methodology
[4]	2020	42	2015–2019	Alzheimer's, Parkinson's, and Schizophrenia
[5]	2017	300	1995–2017	Image classification and lesion segmentation
[6]	2019	65	2008–2018	Schizophrenia, ASD, Parkinson's, and Epilepsy
[7]	2017	85	–	Models/Algorithms, Alzheimer's, etc.

The objective of this project is to develop a classification model (or several) that can identify the presence or absence of tumors in brain magnetic resonance images, with the help of various pre-trained neural networks, the application of fine-tuning, and Ensemble learning (combination of several models). The model will be trained using an MRI dataset and validated to ensure its accuracy and effectiveness [8]. The implementation of this system may represent an important advancement in the automation of medical diagnosis, with the

potential to improve diagnostic accuracy and reduce the time required for decision-making.

II. METHODOLOGY

The present methodology aims to describe the process followed for the development and evaluation of classification models for identifying tumors in brain magnetic resonance images (Brain MRI). This project used three pre-trained convolutional neural network architectures (VGG16, Xception, and InceptionResNetV2) with fine-tuning and subsequent integration through Ensemble learning to improve classification accuracy (see Fig. 1 for the complete pipeline).

In order to identify/explain the model's predictions in the current methodology, the use of the Grad-Cam (XAI) technique was also integrated, which consists of grounding/explaining the models' decision (on which areas of the image the model bases its decision).

A. Data Preparation

In this project, the Mateusz Buda LGG MRI Segmentation dataset [8] available on Kaggle was used. The download was automated through the kagglehub library, ensuring fast, consistent, and organized retrieval of files. This dataset is quite versatile, allowing different types of work.

The dataset is organized into 110 folders, each corresponding to a specific patient case and named according to a unique Case ID that identifies the originating institution. Each folder contains MRI images following the naming convention: TCGA_<inst>_<patient>_<slice>.tif. Given that each folder contains a variable number of slices per patient, this structured naming ensures precise identification of every image.

The dataset consists of brain MRI scans and their corresponding segmented masks. Original images are identified by filenames without the `_mask` suffix, whereas the associated masks are explicitly labeled with the `_mask` suffix. To manage the relationship between these files, a DataFrame-based organization was implemented, comprising the following structure:

TABLE II
DATAFRAME STRUCTURE FOR IMAGE-MASK MAPPING

Column Name	Description
id	Unique identifier.
image_path	Absolute or relative path to the MRI slice.
mask_path	Corresponding path to the segmented mask.
diagnosis (tumour)	Label indicating the presence or absence of a tumor.

Subsequently, a custom image processing function was implemented to check for the presence of tumors in the masks (diagnosis). Simply put, the function analyzed the pixel values in each mask, values greater than zero indicated the presence of a tumor. This information was added as a new column called 'tumour', with the labels 'positive' and 'negative' representing, respectively, the presence or absence of a tumor in that patient.

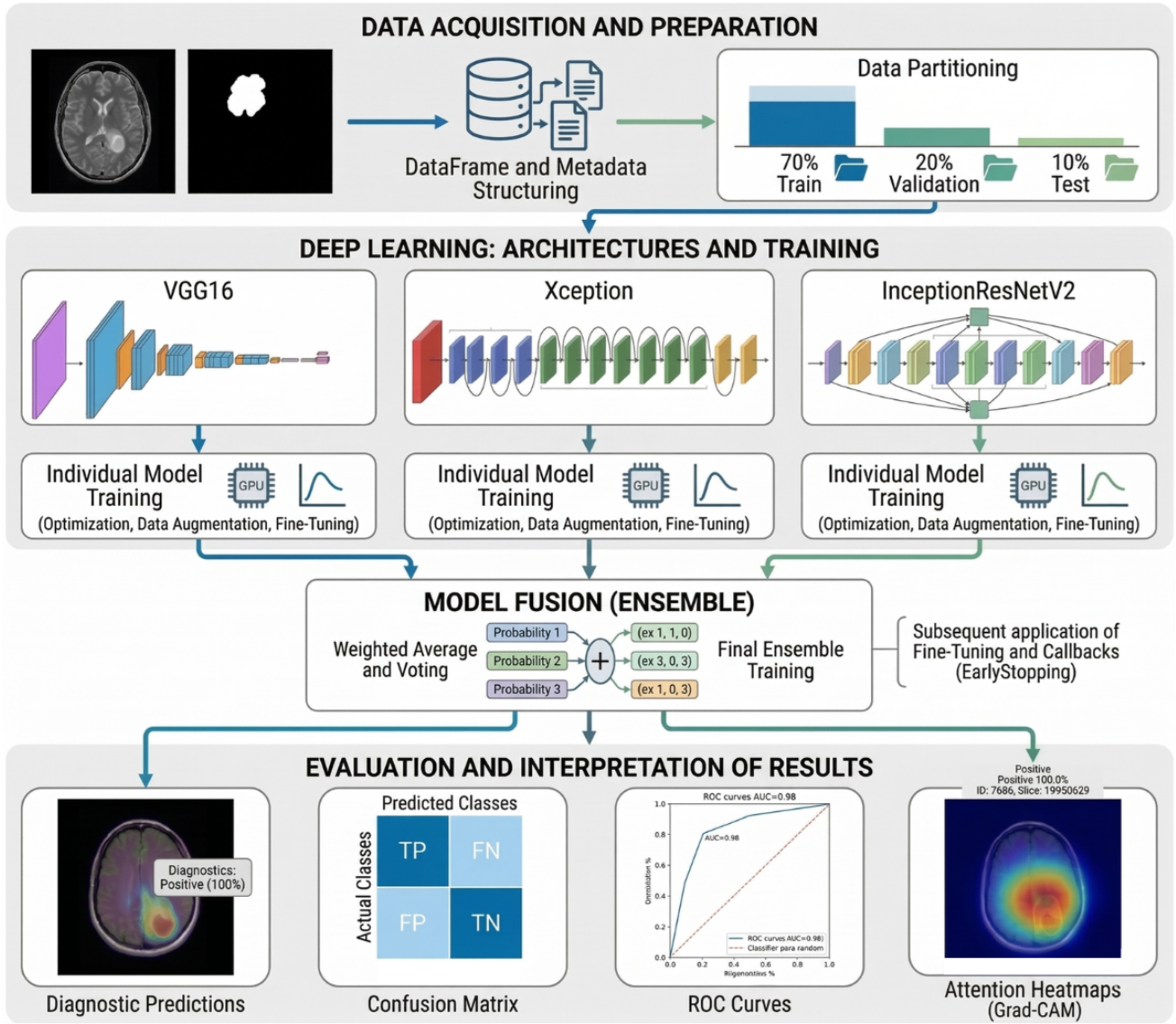


Fig. 1. Overview of the proposed methodology. The workflow starts with raw data preparation, followed by data partitioning, definition of CNN models (VGG16, Xception, and InceptionResNetV2), ensemble training with callbacks, and finally, model evaluation using various metrics and XAI technique.

B. Data Splitting

The dataset was partitioned into training, validation, and test sets using a stratified approach to ensure class balance across all subsets. This process was implemented through the `train_test_split` function in two stages, initially separating the data into training and test sets, and subsequently dividing the training portion to create the validation set. The final distribution of the data was as follows:

- **Training Set:** 70% of the total data;
- **Validation Set:** 20% of the total data;
- **Test Set:** 10% of the total data.

To optimize the learning process, all images were normalized and resized to meet the input requirements of the selected architectures. Additionally, data augmentation techniques were applied to the training set to enhance generalization and

mitigate overfitting.

C. Model Architectures

For the classification task, three state-of-the-art pre-trained architectures were selected due to their proven effectiveness in computer vision:

- **VGG16:** A classic architecture that uses a stack of convolutions of small filters, followed by pooling layers. It is known for its simplicity and ability to capture hierarchical features in images [9]. For this project, the final dense layers were removed and replaced by fully connected layers. This process was defined through generic customizable model creation and training functions in the project code.
- **Xception:** Based on the concept of depthwise separable convolutions, this architecture is an evolution of Inception, designed to increase computational efficiency

while maintaining high accuracy [10]. Its design allows capturing complex patterns in medical images, making it suitable for the task of identifying brain tumors.

- **InceptionResNetV2:** Combines Inception modules with residual connections, allowing the learning of even more robust representations [11]. This architecture is capable of balancing depth and computational efficiency, which was crucial for dealing with the visual and structural variability of tumors in the images.

The models were initialized with pre-trained weights on ImageNet, allowing the leveraging of previously learned generic features. Furthermore, fine-tuning was also applied (unfreezing some of the base model layers) in order to specialize the models in the medical problem in question. The final layers of each model were adapted by adding dense layers with the ReLu activation function and Dropout layers to avoid overfitting.

In order to make a fair later comparison, all models followed the same pattern, except the base part of each one, through the creation of an initial custom function that creates the model and then another function to perform the training.

D. Training and Validation of the Models

The training of each of the models was carried out using the training and validation sets, the latter serves to evaluate the training performance of the model, helping to measure the model's ability to generalize to data not seen in the training set, verifying whether or not overfitting occurred.

For each model (VGG16, Xception, and InceptionResNetV2), the following steps were followed.

1) *Configuration of Hyperparameters, regularization strategies, and callbacks:* The Loss function used was categorical `cross_entropy`, given that the labels were also encoded in the one-hot format. This approach allowed the models to evaluate probabilities for each class more explicitly.

The Adam optimizer was chosen due to its adaptability and effectiveness in adjusting weights during training.

Regarding regularization, dropout layers were added between dense layers to reduce the possibility of overfitting. Due to the uniform distribution of classes in the training set, the weights of each class were balanced so that the models did not favor the majority classes, which was the class corresponding to the case of no tumor existing.

Callbacks like EarlyStopping were implemented to prevent overfitting and optimize the performance of each model. EarlyStopping stops training when performance on the validation set does not improve for a predefined number of epochs (patience)

2) *Training:* In a first phase, each model was trained for 10 epochs, a quite small number due to the long training time and available computational capacity. Subsequently, fine-tuning was applied in which some layers of the base model were unfrozen, and each model was trained for 24 epochs with each model stopping between 9 and 15 epochs. In this way, training results improved significantly.

Training was monitored by the validation metric `val_accuracy`, and graphs of the training curves (accuracy and loss) were generated to evaluate progress (Fig. 2).

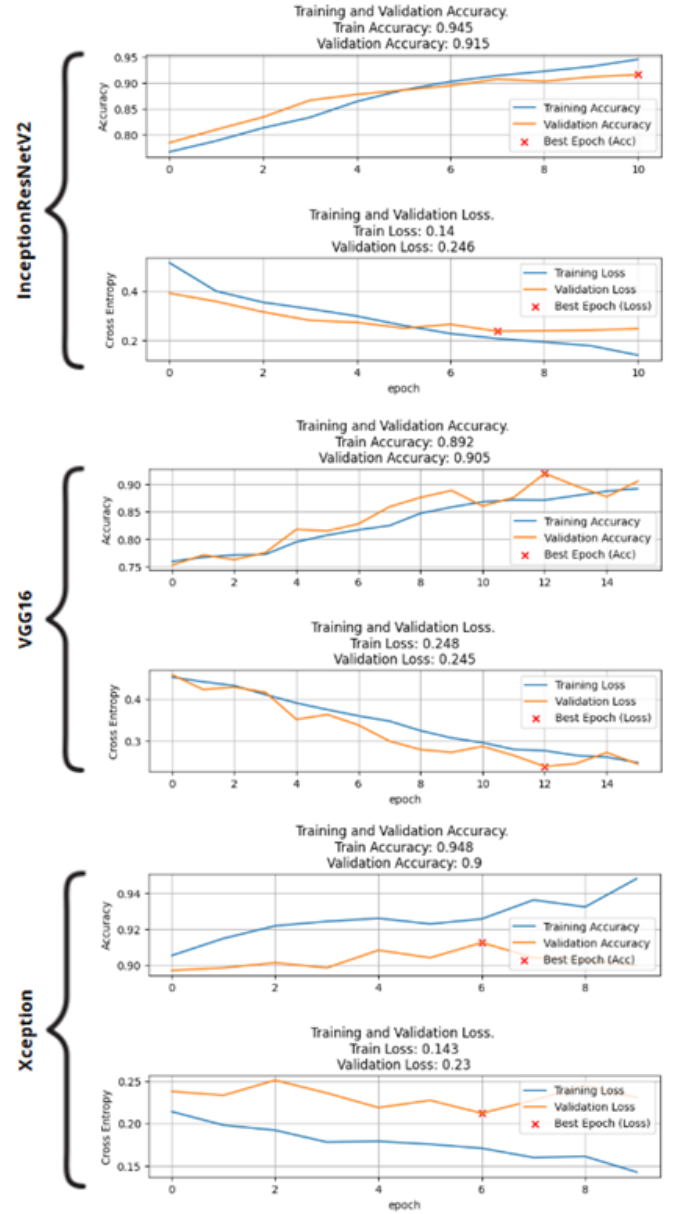


Fig. 2. Training and validation performance curves for the selected models (InceptionResNetV2, VGG16, and Xception). The top graphs display Accuracy, and the bottom graphs display Cross Entropy Loss over epochs.

It is worth remembering that each model went through adjustments in the final layers, adjustments that were identical for each model, in order to make fair future comparisons.

3) *Validation:* The validation set was used to monitor the ability of models to generalize to data not seen in the training set (Fig. 5). Thus, during training, loss and accuracy metrics in the validation set were analyzed at the end of each epoch, and the model with the best validation performance was saved for subsequent evaluation on the test set.

E. Ensemble Learning

To improve the robustness and general performance of the classification, an ensemble learning approach was adopted. This technique combines individual predictions from the three trained models (VGG16, Xception, and InceptionResNetV2), taking advantage of the complementarity of the different architectures.

The combination of predictions was performed using the Voting Ensemble technique, more specifically an arithmetic mean of the predictions of the three models. For each input sample, the class probabilities generated by each model were equally weighted and then summed to obtain the final prediction. The class with the highest final probability was chosen as the ensemble prediction.

This approach allows the individual strengths of each model to contribute to the result, reducing errors that a specific model might make due to its particular limitations.

F. Interpretation with Grad-CAM (XAI)

Explainable artificial intelligence (XAI) methods make AI processes and results (black-boxes) understandable to humans. There are different XAI techniques, such as the Grad-CAM technique.

With the objective of providing greater interpretability to the predictions, the Gradient-weighted Class Activation Mapping (Grad-CAM) technique was used. This technique allowed the visualization of regions of the input images that contributed most to the model's decisions, highlighting relevant areas associated with the presence of tumors.

The Grad-CAM analysis was conducted on examples from the test set, generating heatmaps overlaid on the original images. These visualizations were evaluated to verify the consistency of the predictions with the typical visual characteristics of brain tumors. Furthermore, the results were compared qualitatively for each model, reinforcing the reliability of the system.

G. Performance Evaluation

The performance evaluation of the models was carried out on the test set, using various metrics to obtain a complete analysis of the models' and the ensemble's effectiveness.

Each metric was chosen to provide a specific view on the quality of predictions.

1) *Accuracy*: Accuracy measures the proportion of correct predictions relative to the total samples evaluated.

$$\text{Accuracy} = \frac{\text{Number of correct predictions (TP + TN)}}{\text{Total number of samples (TP + TN + FP + FN)}}$$

This metric provides an overview of performance, but it can be misleading in cases of imbalance, as a high proportion of one class can inflate the accuracy value. However, the fact that an initial balancing of class weights was done reduced this tendency, making the accuracy more representative of actual performance.

2) *Area Under the ROC Curve (AUC)*: The AUC metric measures the model's ability to discriminate between positive and negative classes, evaluating all possible threshold configurations. The ROC curves show the relationship between the true positive rate (sensitivity) and the false positive rate (1 - specificity) (Fig. 3).

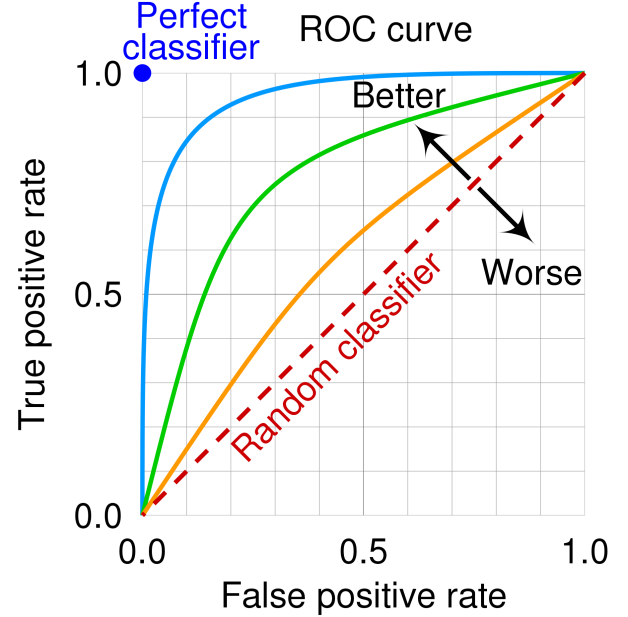


Fig. 3. ROC curve analysis. The diagonal dashed red line represents a random classifier. Curves moving towards the top-left corner (blue and green lines) indicate better performance, while those close to the diagonal (orange line) show worse performance.

An AUC close to 1 (top left corner) indicates an excellent ability to separate the classes, that is, an excellent ability to classify a given class. While an AUC close to 0.5 indicates a performance similar to random classification.

This metric is particularly useful in medical problems, as it evaluates how the model balances false positives and false negatives at different decision thresholds.

3) *Confusion Matrix*: The confusion matrix (Fig. 4) provides a detailed representation of the model's predictions in terms of hits and misses, categorized into four fundamental metrics:

- **True Positive (TP)**: Tumors correctly identified.
- **False Positive (FP)**: Non-tumor cases incorrectly classified as tumors.
- **True Negative (TN)**: Absence of tumor correctly identified.
- **False Negative (FN)**: Tumors not detected by the model.

Based on the values from the confusion matrix, additional performance metrics are calculated to provide a more granular evaluation:

- **Precision**: Measures the proportion of true positive predictions relative to the total number of positive predictions. This metric is particularly relevant in medical

		Predicted	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

Fig. 4. Confusion Matrix.

diagnostics where false positives can lead to unnecessary costs and patient anxiety.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):** Evaluates the proportion of actual positive cases that were correctly identified. In the context of tumor detection, this is a critical metric to avoid false negatives, where a failure to detect a tumor can be life-threatening.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-score:** The harmonic mean between Precision and Recall. It is an essential metric for scenarios with class imbalance, providing a balanced view of both precision and sensitivity.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

III. RESULTS AND DISCUSSION

The results presented in this section highlight the performance of trained models, including VGG16, Xception, InceptionResNetV2 architectures, and an Ensemble learning approach. Metrics such as accuracy, ROC curves (AUC), confusion matrix, precision, recall, and F1-score were analyzed, alongside the qualitative interpretation of predictions made by each model using the XAI (Explainable artificial intelligence) Grad-CAM technique.

The discussion covers the strengths and limitations of each model individually, as well as the positive impact of combining the models into the Ensemble, emphasizing the applicability of proposed methods in clinical contexts, such as detecting brain tumors in magnetic resonance images.

A. Performance of Individual Models (VGG16, Xception, InceptionResNetV2)

The results for each model were evaluated on the test set and are presented in Table III. In a first phase, where fine-tuning was not applied to any of the 3 models, it is observed that the

Xception model presented the highest accuracy, followed by InceptionResNetV2 and finally VGG16.

The AUC metric was applied only at the end, as it makes more sense to use it then, since we will obtain models with even higher performance through the application of fine-tuning.

Analysis of the confusion matrix revealed that Xception had the lowest false negative rate, which is particularly relevant in medical problems, since failing to identify a tumor can have serious consequences, regardless of the patient's condition. On the other hand, VGG16 presented the highest false positive rate, indicating a greater propensity to misclassify negative images as positive.

TABLE III
MODEL PERFORMANCE ON TEST SET (BEFORE FINE-TUNING)

Model	Accuracy	Recall	Precision	F1-score	AUC
InceptionResNetV2	0.78	0.79	0.77	0.77	–
VGG16	0.75	0.74	0.73	0.73	–
Xception	0.83	0.85	0.82	0.82	–

Subsequently, fine-tuning was applied to each model, and performance improved greatly at all levels, as can be seen in Table IV.

TABLE IV
MODEL PERFORMANCE ON TEST SET (AFTER FINE-TUNING)

Model	Accuracy	Recall	Precision	F1-score	AUC
InceptionResNetV2	0.88	0.86	0.87	0.87	0.952
VGG16	0.90	0.89	0.88	0.89	0.961
Xception	0.91	0.89	0.89	0.90	0.968

Thus, the performance of the three models increased overall, establishing them as the final models.

The AUC metric reflects the models' ability to discriminate between classes consistently, with all values above 0.95, indicating excellent separation between the positive and negative classes (Fig. 5).

Accuracy, combined with the F1-score, suggests that models not only correctly classify a significant proportion of samples but also maintain a balance across classes, which is essential in clinical contexts.

The Xception model performed best, achieving an AUC of 0.968, the highest among the models, and an F1-score of 0.90, indicating a good balance between Precision and Recall.

The VGG16 model showed a slight inferiority in terms of AUC (0.961), but demonstrated good precision nonetheless (0.88) and maintains a high F1-score (0.89).

As for the InceptionResNetV2 model, it also exhibited robust performance, with an AUC of 0.952 and an F1-score of 0.87, evidencing its classification capacity, although it was the weakest of the three.

These results reinforce the effectiveness of tested models for medical image classification tasks, with Xception demonstrating a slight advantage in terms of generalization and accuracy,

making it a preferred choice for sensitive applications like tumor detection.

B. Performance of Ensemble learning

The Ensemble learning approach combined the three models through an average voting of probabilities. The Ensemble's performance surpassed that of individual models in terms of accuracy and AUC, as shown in Table V and Fig. 5.

TABLE V
PERFORMANCE OF THE ENSEMBLE ON THE TEST SET

Model	Accuracy	Recall	Precision	F1-score	AUC
Ensemble	0.93	0.92	0.92	0.92	0.979

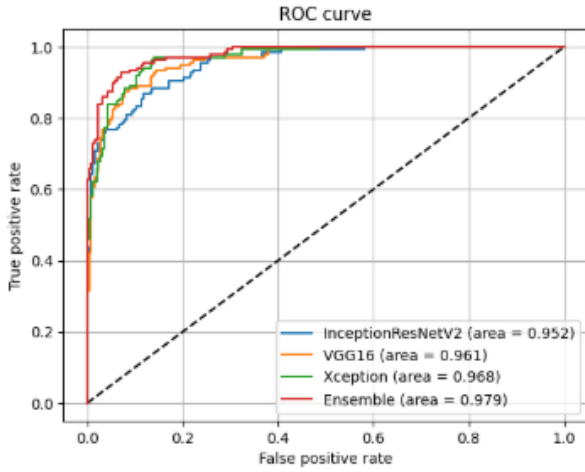


Fig. 5. Final ROC curve comparison on the test set for all evaluated models. The Ensemble approach achieved the highest area under the curve (AUC = 0.979), outperforming the individual architectures: Xception (AUC = 0.968), VGG16 (AUC = 0.961), and InceptionResNetV2 (AUC = 0.952). The black dashed line represents the performance of a random classifier.

The Ensemble benefited from the complementarity among the models' architectures, reducing the error rate and achieving robust generalization capacity.

As already stated, all models presented an AUC value above 0.95. The Xception model stood out with the highest AUC among the 3 models, reflecting superior generalization compared to the other models. In turn, the Ensemble achieved the highest AUC (0.979), consolidating its effectiveness by combining the individual models' predictions.

The Ensemble's ROC curve reveals a true positive rate consistently superior compared to the individual models' curves, especially at thresholds where false positives would be a critical factor. This characteristic is particularly advantageous in medical contexts, where minimizing false negatives is essential to avoid medical diagnostic failures.

The Ensemble's confusion matrix revealed a significant reduction in false negatives compared to individual models, which reinforces its application in clinical contexts (Fig. 6).

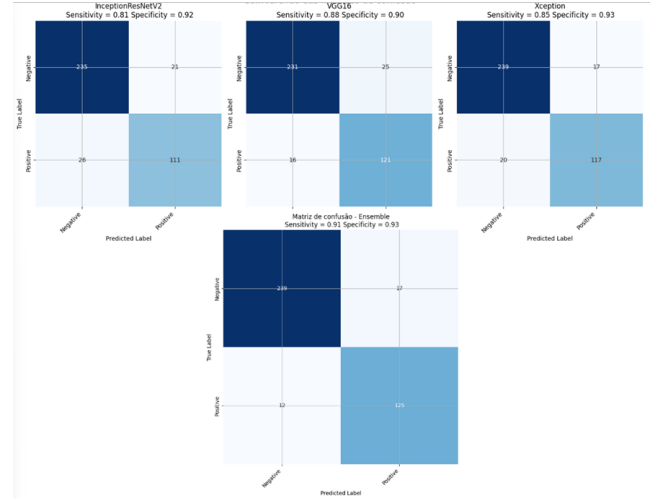


Fig. 6. Comparison of confusion matrices for the evaluated models (Inception-ResNetV2, VGG16, Xception) and the final Ensemble approach. The matrices display the counts of true negatives (top-left), false positives (top-right), false negatives (bottom-left), and true positives (bottom-right). The Sensitivity and Specificity values for each model are also provided.

C. Interpretation with Grad-CAM

The Grad-CAM technique was applied to test set examples to interpret the models' decisions. The generated heatmaps showed that the most relevant regions for predictions were consistently associated with areas where tumors were present, qualitatively validating the models' predictions (Fig. 7, 8 and 9).

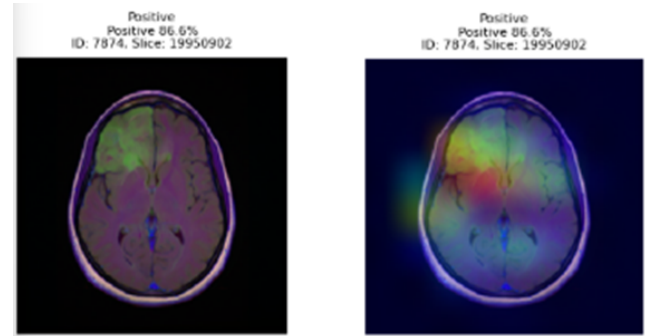


Fig. 7. Feature importance analysis for the evaluated model (InceptionRes-NetV2) and the final Ensemble approach. Heatmaps overlaid on MRI slices indicate the regions of highest contribution to the model's decision.

The visual interpretation offered by Grad-CAM also allowed identifying error cases, such as false positives and negatives, where models highlighted (or based their decision on) less relevant regions or artifacts unrelated to the Tumor. This suggests that further improvements could be achieved with adjustments to final layers or other types of more refined Data Augmentation techniques

D. Summary

The results show that the architectures used, especially Xception, presented remarkable performance in classifying

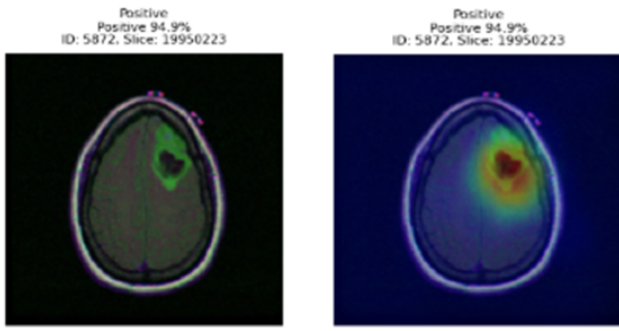


Fig. 8. Feature importance analysis for the evaluated model (VGG16) and the final Ensemble approach. Heatmaps overlaid on MRI slices indicate the regions of highest contribution to the model's decision.

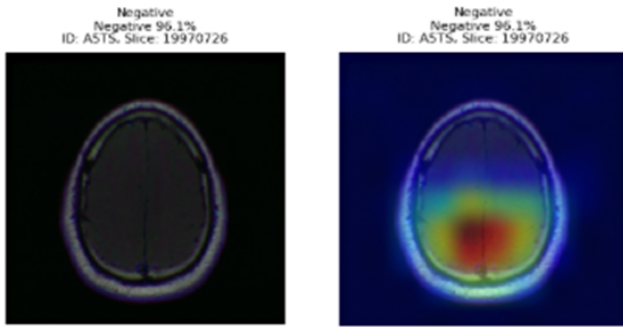


Fig. 9. Feature importance analysis for the evaluated model (Xception) and the final Ensemble approach. Heatmaps overlaid on MRI slices indicate the regions of highest contribution to the model's decision.

brain tumors in MRI images. The Ensemble approach proved effective by combining the strengths of each individual model, achieving superior and more reliable results.

Using Grad-CAM contributed significantly to model interpretation, increasing confidence in predictions and offering insights into areas for improvement. However, some challenges remain, such as the need to further reduce the number of false positives and improve accuracy in more ambiguous cases.

Overall, the results obtained reinforce the potential of convolutional neural networks and Ensemble techniques for clinical applications, especially in brain tumor detection, where accuracy is essential for medical decision-making support.

IV. CONCLUSION

In this work, different approaches based on convolutional neural networks (VGG16, Xception, and InceptionResNetV2) and an Ensemble learning technique for binary classification of brain tumors in magnetic resonance imaging (MRI) were explored. The results highlighted the effectiveness of these architectures in the binary classification task, with high performance metrics, such as accuracy, F1-score, and AUC, reflecting a strong generalization capacity.

The Xception model proved to be the best individual solution, achieving the highest AUC and lowest false positive rate, critical characteristics in clinical contexts. However, Ensemble learning demonstrated to be the most robust approach, as

expected, since it combines the strengths of the three models, offering superior overall performance with 93 % accuracy and a 0.98 AUC.

Using the Grad-CAM technique allowed interpreting the models' predictions, identifying the most relevant areas for decisions, and highlighting its practical applicability in AI-assisted diagnoses. Despite promising results, challenges like reducing false positives/negatives and classifying more ambiguous cases remain opportunities for future work.

In conclusion, this study confirms the potential of deep neural networks and Ensemble techniques to support the early detection of brain tumors, providing a reliable and effective tool to complement medical decision-making. Future investigations could explore increasing data diversity, more advanced pre-processing techniques, and even more specialized architectures to refine the results obtained.

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